

Muhammad Farzaib Khan

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Professional Summary

Software Engineer with 4+ years of experience (2.5+ years full-time, plus part-time experience as a Working Student) building enterprise software across the full stack, with hands-on exposure to applying AI/ML in production systems. Skilled in RESTful API development, cloud-native microservices on Azure and AWS, and large-scale system modernization, including a zero-downtime migration from .NET Framework 4.8 to .NET 9. Experienced designing and deploying MCP (Model Context Protocol) agents and ML-based forecasting features, alongside modern frontend frameworks (Blazor, Angular, React) and backend technologies (C#, .NET, ASP.NET Core).

Work Experience

Software Engineer (Working Student)

Sep 2024 – Jul 2026

ATRON Group (DDS-Digital Data Systems GmbH), Germany

- Designed and built MCP (Model Context Protocol) agents integrating AI into internal tooling, reducing the steps required to search for information from 3+ down to 1
- Replaced manual, error-prone document workflows with RESTful APIs, cutting client-reported processing errors by approximately 75% while processing 500+ documents per month
- Built responsive, component-based web applications using Blazor (.NET 9) for 30+ internal users, replacing legacy desktop tooling
- Took ownership of a legacy Windows Forms application, restructured it using the MVVM pattern, and migrated it from .NET Framework 4.8 to .NET 9 with zero downtime
- Added ML-based forecasting to production dashboards, generating 20 client-specific metrics and visualizations to support transport-related decision-making
- Implemented OIDC-based authentication across 3 identity providers (Microsoft Identity, client-owned, and product-owned identity systems), supporting the organization's ISO 27001 and GDPR compliance requirements

Software Engineer

Aug 2021 – Apr 2024

NetSol Technologies, Lahore, Pakistan

- Built enterprise frameworks in C#, .NET Core, ASP.NET Core, and Entity Framework, improving development efficiency by 30%
- Developed framework modules for localization, caching, and validation, and led migration of the core framework from .NET Framework 4.8 to .NET 8
- Implemented OIDC-based authentication and authorization across multiple client integrations, and built WPF desktop tooling using the MVVM pattern
- Built a Windows Service using SQL Server Service Broker External Activator (SSBEA) to audit and log all database changes, capturing change data and persisting it for traceability and compliance
- Converted a .NET WPF desktop module to an Angular and NgRx frontend, replicating MVVM state management in a reactive, testable architecture
- Architected and deployed cloud-native microservices using Docker, Kubernetes, Azure, AWS, and CI/CD pipelines, and deployed and maintained web applications on IIS (Windows Server)
- Led Agile technical design discussions and code reviews, reducing project delays by 20%
- Conducted 15+ technical interviews and built onboarding programs, reducing new-hire ramp-up time by 30%

Software Engineer Trainee

May 2021 – Jul 2021

NetSol Technologies, Lahore, Pakistan

- Analyzed business and technical domains related to leasing and financing services
- Collaborated in hands-on training to design and build custom enterprise frameworks

Intern

Jul 2019 – Aug 2019

Pakistan Civil Aviation Authority, Lahore, Pakistan

- Developed an inventory management system for the organization

Projects

StreamPulse

2026

Personal project | <https://github.com/jrvs98/StreamPulse>

- Built a C++17 streaming-quality monitor that measures frame latency, dropped frames, read errors, bitrate, and variance, with live JSON metrics in an HTTP dashboard
- Implemented synthetic stream monitoring through the /api/metrics endpoint and validated its metrics with an automated end-to-end test
- Tech stack: C++17, CMake, OpenCV, POSIX sockets, HTML/CSS/JavaScript, CTest

Hybrid RAG Service

2026

Personal project | <https://github.com/jrvs98/hybrid-rag>

- Prototyped a FastAPI RAG service combining BM25 keyword search and dense vector search with reranking, cited answer generation, and automated citation verification
- Implemented in-memory BM25 and cosine-similarity retrieval, backed by 29 passing automated tests
- Tech stack: Python, FastAPI, Pydantic, pytest, Ruff

Prompt Delta Eval

2026

Personal project | <https://github.com/jrvs98/prompt-delta-eval>

- Built an evaluation framework measuring whether prompt engineering still improves LLM output quality across bare, minimal, and engineered prompt conditions
- Implemented a provider-agnostic architecture supporting OpenAI, Anthropic, Google, and local Ollama models, plus a mock provider for pipeline testing
- Built automated and LLM-judged scoring with a separate format-compliance check, aggregating results into per-model, per-task HTML reports
- Tech stack: Python, SQLite, pytest, GitHub Actions CI

Coding Agent Evaluation Harness

2026

Personal project | <https://github.com/jrvs98/CodingAgentEval>

- Built a Planner/Coder/Critic coding agent and an 18-task benchmark across 3 Python repositories, with hidden grading tests for bug fixes, features, and multi-file changes
- Implemented single-agent and multi-agent runners with JSONL reports for pass rate, cost, latency, iterations, and failure reasons
- Added 6 adversarial tasks, patch-policy checks, and cost/iteration limits; the recorded Gemini multi-agent adversarial run passed 5/6 tasks (83.33%)
- Tech stack: Python, JSONL, unittest, subprocess, Anthropic/Gemini/GitHub Models APIs

Team Policy Assistant

2026

Personal project | <https://github.com/jrvs98/TeamPolicyAssistant>

- Built a role-protected policy Q&A application with FastAPI, React/TypeScript, and Keycloak; administrators upload policies while employees search, read cited answers, and submit feedback
- Implemented asynchronous PDF/Markdown ingestion through RabbitMQ, including Tesseract OCR fallback, text chunking, and pgvector-backed retrieval in PostgreSQL
- Added extractive answers with source citations and fallback for weak or missing evidence, plus admin evaluations of retrieval and refusal behavior
- Tech stack: Python, FastAPI, React, TypeScript, PostgreSQL/pgvector, RabbitMQ, Keycloak, Docker Compose, Tesseract OCR

Education

Hochschule Fulda

Apr 2024 – Expected 2027

Master of Science in Global Software Development, Fulda, Germany

National University of Computer & Emerging Sciences (FAST-NU)

Sep 2017 – Sep 2021

Bachelor of Science in Computer Science, Lahore, Pakistan

Skills

Programming: C#, C++, Python, TypeScript, JavaScript, SQL/T-SQL

Backend & Data: .NET 9, .NET Core, ASP.NET Core, Entity Framework, FastAPI, RESTful APIs, microservices, SQL Server, PostgreSQL/pgvector, MongoDB, Elasticsearch

Frontend: React/Redux Toolkit, Angular/NgRx, Blazor, HTML/CSS, WPF, Windows Forms

Cloud & Delivery: Azure, AWS, Azure DevOps, Docker, Docker Compose, Kubernetes, CI/CD, Git, GitLab, RabbitMQ

AI & Security: Machine learning, forecasting, LLM API integration, retrieval-augmented search, Tesseract OCR, Model Context Protocol (MCP), OIDC, Keycloak

AI Tools: Codex, Copilot, Cowork

Methods & Languages: MVVM, SOLID, Agile/Scrum; English (fluent), German (B1), Urdu/Hindi (native)

Certifications

Microsoft Certified: Azure Fundamentals; Machine Learning Fundamentals with Python; Data Scientist with Python; Data Analyst with Python; Certified Microsoft Innovative Educator.